

LXP32

a lightweight open source 32-bit CPU core

Technical Reference Manual

Version 1.5.1

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Chapter 1

Introduction

1.1 Main features

LXP32 (*Lightweight eXecution Pipeline*) is a small 32-bit CPU IP core optimized for FPGA implementation. Its key features include:

- portability (described in behavioral VHDL-93, not tied to any particular vendor);
- 3-stage hazard-free pipeline;
- 256 registers implemented as a RAM block;
- a simple instruction set with only 30 distinct opcodes;
- separate instruction and data buses, optional instruction cache;
- WISHBONE compatibility;
- 8 interrupts with hardwired priorities;
- optional divider.

As a lightweight CPU core, LXP32 lacks some features of more advanced processors, such as nested interrupt handling, debugging support, floating-point and memory management units. LXP32 is based on an original ISA (Instruction Set Architecture) which does not currently have a C compiler. It can be programmed in the assembly language covered by Appendix C.

Two major hardware versions of the CPU are provided: LXP32U which does not include an instruction cache and uses the Low Latency Interface (Section 3.5) to fetch instructions, and LXP32C which fetches instructions over a cached WISHBONE bus protocol. These versions are otherwise identical and have the same instruction set architecture.

1.2 Implementation estimates

Typical results of LXP32 core FPGA implementation are presented in Table 1.1. Note that these data are only useful as rough estimates, since actual results depend greatly on tool versions and configuration, design constraints, device utilization ratio and other factors.

Data on two configurations are provided:

- *Compact*: LXP32U (without instruction cache), no divider, 2-cycle multiplier.
- *Full*: LXP32C (with instruction cache), divider, 2-cycle multiplier.

The slowest speed grade was used for clock frequency estimation.

Table 1.1: Typical results of LXP32 core FPGA implementation

Resource	Compact	Full
Microsemi® IGLOO®2 M2GL005-FG484		
Logic elements (LUT+DFF)	1457	2086
LUTs	1421	1999
Flip-flops	706	1110
Mathblocks (MACC)	3	3
RAM blocks (RAM1K18)	2	3
Clock frequency	107.7 MHz	109.2 MHz
Xilinx® Artix®-7 xc7a15tfgg484-1		
Slices	235	365
LUTs	666	1011
Flip-flops	528	883
DSP blocks (DSP48E1)	4	4
RAM blocks (RAMB18E1)	2	3
Clock frequency	111.9 MHz	120.2 MHz

1.3 Structure of this manual

General description of the LXP32 operation from a software developer's point of view can be found in Chapter 2, *Instruction set architecture*. Future versions of the LXP32 CPU are intended to be at least backwards compatible with this architecture.

Topics related to hardware, such as synthesis, implementation and interfacing other IP cores, are covered in Chapter 3, *Integration*. A brief description of the LXP32 pipelined architecture is provided in Chapter 4, *Hardware architecture*. The LXP32 IP core package includes a verification environment (self-checking testbench) which can be used to simulate the design as described in Chapter 5, *Simulation*.

Documentation for tools shipped with the LXP32 IP core package (assembler/linker, disassembler and interconnect generator) is provided in Chapter 6, *Development tools*.

Appendices include a detailed description of the LXP32 instruction set, instruction cycle counts and LXP32 assembly language definition. WISHBONE datasheet required by the WISHBONE specification is also provided.

Chapter 2

Instruction set architecture

2.1 Data format

Most LXP32 instructions work with 32-bit data words. A few instructions that address individual bytes use little-endian order, that is, the least significant byte is stored at the lowest address. Signed values are encoded in a 2's complement format.

2.2 Instruction format

All LXP32 instructions are encoded as 32-bit words, with the exception of **lc** (*Load Constant*), which occupies two adjacent 32-bit words. Instructions in memory must be aligned to word boundaries.

Most arithmetic and logical instructions take two source operands and write the result to an independent destination register. General instruction format is presented on Figure 2.1.

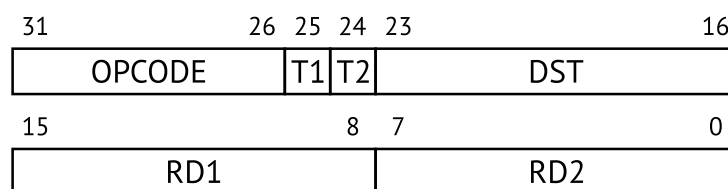


Figure 2.1: LXP32 instruction format

This format includes the following fields:

1. OPCODE – a 6-bit instruction code (see Appendix A).

2. T1 – type of the RD1 field.
3. T2 – type of the RD2 field.
4. DST – register number (usually the destination register).
5. RD1 – register/direct operand 1.
6. RD2 – register/direct operand 2.

Some of these fields may not have meaning for a particular instruction; such unused fields are replaced with zeros.

DST field specifies one of the 256 LXP32 registers. RD1 and RD2 fields can denote either source register operands or direct (immediate) operands: if the corresponding T field is 1, RD value is a register number, otherwise it is interpreted as a direct signed byte in a 2's complement format (valid values range from -128 to 127).

For example, consider the following instruction that adds 10 to r0 and writes the result to r1:

```
add r1, r0, 10
```

In this example, OPCODE is 010000, T1 is 1, T2 is 0, DST is 00000001, RD1 is 00000000 and RD2 is 00001010. Hence, the instruction is encoded as 0x4201000A.

For convenience, some instructions have alias mnemonics. For example, LXP32 does not have a distinct **mov** opcode: instead, **mov** dst, src is an alias for **add** dst, src, 0.

A complete list of LXP32 instructions is provided in Appendix A.

2.3 Registers

LXP32 has 256 registers denoted as r0 – r255. The first 240 of them (from r0 to r239) are general-purpose registers (GPR), the last 16 (from r240 to r255) are special-purpose registers (SPR). For convenience, some special-purpose registers have alias names: for example, r255 can be also referred to as sp (stack pointer). Special purpose registers are listed in Table 2.1. Some of these registers are reserved: the software should not access them.

All registers are zero-initialized during the CPU reset.

Table 2.1: LXP32 special-purpose registers

Alias name	Generic name	Description
iv0	r240	Interrupt vector 0 (Section 2.7)
iv1	r241	Interrupt vector 1 (Section 2.7)
iv2	r242	Interrupt vector 2 (Section 2.7)
iv3	r243	Interrupt vector 3 (Section 2.7)
iv4	r244	Interrupt vector 4 (Section 2.7)
iv5	r245	Interrupt vector 5 (Section 2.7)
iv6	r246	Interrupt vector 6 (Section 2.7)
iv7	r247	Interrupt vector 7 (Section 2.7)
—	r248 – r251	<i>Reserved</i>
cr	r252	Control register (Section 2.7)
irp	r253	Interrupt return pointer (Section 2.7)
rp	r254	Return pointer (Section 2.6)
sp	r255	Stack pointer (Section 2.5)

2.4 Addressing

All addressing in LXP32 is indirect. In order to access a memory location, its address must be stored in a register; any available register can be used for this purpose.

LXP32 uses a 32-bit address space. Each address refers to an individual byte. Some instructions, namely **lsb** (*Load Signed Byte*), **lub** (*Load Unsigned Byte*) and **sb** (*Store Byte*) provide byte-granular access, in which case all 32 bits in the address are significant. Otherwise the least two address bits are ignored as LXP32 doesn't support unaligned access to 32-bit data words (during simulation, a warning is emitted if such a transaction is attempted).

A special rule applies to pointers that refer to instructions: since instructions are always word-aligned, the least significant bit is interpreted as the IRF (*Interrupt Return Flag*). See Section 2.7 for details.

2.5 Stack

The current pointer to the top of the stack is stored in the sp register. To the hardware this register is not different from general purpose registers, that is, in no situation does the CPU access the stack implicitly (procedure calls and interrupts use register-based conventions).

Software can access the stack as follows:

```

// push r0 on the stack
sub sp, sp, 4
sw sp, r0
// pop r0 from the stack
lw r0, sp
add sp, sp, 4

```

Before using the stack, the `sp` register must be set up to point to a valid memory location. The simplest software can operate stackless, or even without data memory altogether if registers are enough to store the program state.

2.6 Calling procedures

Procedure calls are supported in LXP32 by the **xcall** instruction which saves the address of the next instruction in the register specified as the first operand and transfers execution to the address pointed by the second operand. In a common case when the first operand is `rp`, the instruction can be also written as **call** which takes only one operand.

Return from a procedure is performed by the **jmp** instruction; **jmp rp** also has a **ret** alias.

If a procedure must in turn call a nested procedure itself, the return address in the `rp` register will be overwritten by the **call** instruction. Hence, unless it is a tail call (see below), the procedure must save the `rp` value somewhere; the most general solution is to use the stack:

```

sub sp, sp, 4
sw sp, rp
...
lc r0, Nested_proc
call r0 // equivalent to xcall rp, r0
...
lw rp, sp
add sp, sp, 4
ret // equivalent to jmp rp

```

Procedures that don't use the **call** instruction (sometimes called *leaf procedures*) don't need to save the `rp` value.

Since **ret** is just an alias for **jmp rp**, one can also use *Compare and Jump* instructions (**cjmpxxx**) to perform a conditional procedure return. For example, consider the following procedure which calculates the absolute value of `r1`:

```

Abs_proc:
    cjmpsge rp, r1, 0 // return immediately if r1>=0
    neg r1, r1 // otherwise, negate r1
    ret // jmp rp

```

A *tail call* is a special type of procedure call where the calling procedure calls a nested procedure as the last action before return. In such cases the **call** instruction can be replaced with **jmp**, so that when the nested procedure executes **ret**, it returns directly to the caller's parent procedure.

Although the LXP32 architecture doesn't mandate any particular calling convention, some general recommendations are presented below:

1. Pass arguments and return values through the r1–r31 registers (a procedure can have multiple return values).
2. If necessary, the r0 register can be used to load the procedure address.
3. Designate r0–r31 registers as *caller-saved*, that is, they are not guaranteed to be preserved during procedure calls and must be saved by the caller if needed. The procedure can use them for any purpose, regardless of whether they are used to pass arguments and/or return values.

2.7 Interrupt handling

Control register

LXP32 supports 8 interrupts with hardwired priority levels (interrupts with lower vector numbers have higher priority). Interrupt vectors (pointers to interrupt handlers) are stored in the iv0–iv7 registers. Interrupt handling is controlled by the cr register (Table 2.2).

Disabled interrupts are ignored altogether: if the CPU receives an interrupt request signal while the corresponding interrupt is disabled, the interrupt will not be processed even if it is enabled later.

Wake-up flag marks the interrupt as a *wake-up interrupt* (see below).

Level-sensitive and invert flags control trigger conditions (Table 2.3).

Like other registers, cr is zero-initialized during the CPU reset, meaning that no interrupts are initially enabled.

Invoking interrupt handlers

Interrupt handlers are invoked by the CPU similarly to procedures (Section 2.6), the difference being that in this case return address is stored in the

Table 2.2: Control register

Bit	Description
0	Enable interrupt 0
...	...
7	Enable interrupt 7
8	Interrupt 0 wake-up flag
...	...
15	Interrupt 7 wake-up flag
16	Interrupt 0 level-sensitive flag
...	...
23	Interrupt 7 level-sensitive flag
24	Interrupt 0 invert flag
...	...
31	Interrupt 7 invert flag

Table 2.3: Interrupt trigger conditions

Level-sensitive	Invert	Trigger condition
0	0	Rising edge
0	1	Falling edge
1	0	High level
1	1	Low level

`irp` register (as opposed to `rp`), and the least significant bit of the register (IRF – *Interrupt Return Flag*) is set.

An interrupt handler returns using the **jmp** `irp` instruction which also has an **iret** alias. Until the interrupt handler returns, the CPU will defer further interrupt processing (although incoming interrupt requests will still be registered). This also means that the `irp` register value will not be unexpectedly overwritten. When executing the **jmp** `irp` instruction, the CPU will recognize the IRF flag and resume interrupt processing as usual. It is also possible to perform a conditional return from the interrupt handler, similarly to the technique described in Section 2.6 for conditional procedure returns.

Wake-up interrupts

When a wake-up interrupt is received, the interrupt handler is not called, but the CPU still resumes execution if halted by the **hlt** instruction. The

effect is similar to invoking an interrupt with an empty handler (containing only **iret**), but without the overhead of interrupt processing. Wake-up interrupts do not affect the CPU when it is not halted.

Unlike normal interrupts, wake-up interrupts are processed even when the CPU executes an interrupt handler for another interrupt.

Non-returnable interrupts

If an interrupt vector has the least significant bit (IRF) set, the CPU will resume interrupt processing immediately. One should not try to invoke **iret** from such a handler since the `irp` register could have been overwritten by another interrupt. This technique can be useful when the CPU's only task is to process external events:

```
// Set the IRF to mark the interrupt as non-returnable
    lc iv0, main_loop@1
    mov cr, 1 // enable the interrupt
    hlt // wait for an interrupt request
main_loop:
// Process the event...
    hlt // wait for the next interrupt request
```

Note that **iret** is never called in this example.

Chapter 3

Integration

3.1 Overview

The LXP32 IP core is delivered in a form of a synthesizable RTL description expressed in VHDL-93. It does not use any technology specific primitives and should work out of the box with major FPGA synthesis software. LXP32 can be integrated in both VHDL and Verilog® based SoC designs.

Major LXP32 hardware versions have separate top-level design units:

- `lxp32u_top` – LXP32U (without instruction cache),
- `lxp32c_top` – LXP32C (with instruction cache).

A high level block diagram of the CPU is presented on Figure 3.1. Schematic symbols for LXP32U and LXP32C are shown on Figure 3.2.

LXP32U uses the Low Latency Interface (LLI) described in Section 3.5 to fetch instructions. This interface is designed to interact with low latency on-chip peripherals such as RAM blocks. It works best with slaves that can return the instruction on the next cycle after its address has been set, although the slave can still introduce wait states if needed. Low Latency Interface can be also connected to a custom (external) instruction cache.

To achieve the least possible latency, some LLI outputs are not registered. For this reason the LLI is not suitable for interaction with off-chip peripherals.

LXP32C is designed to work with high latency memory controllers and uses a simple instruction cache based on a ring buffer. The instructions are fetched over the WISHBONE instruction bus. To maximize throughput, the CPU makes use of the WISHBONE registered feedback signals [CTI_OO] and [BTE_OO]. All outputs on this bus are registered. This version is also

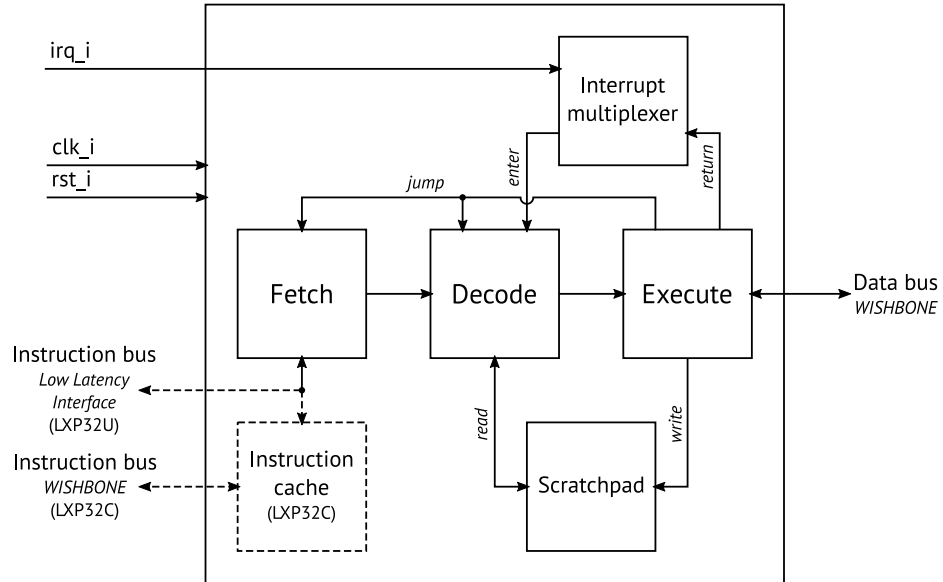


Figure 3.1: LXP32 CPU block diagram

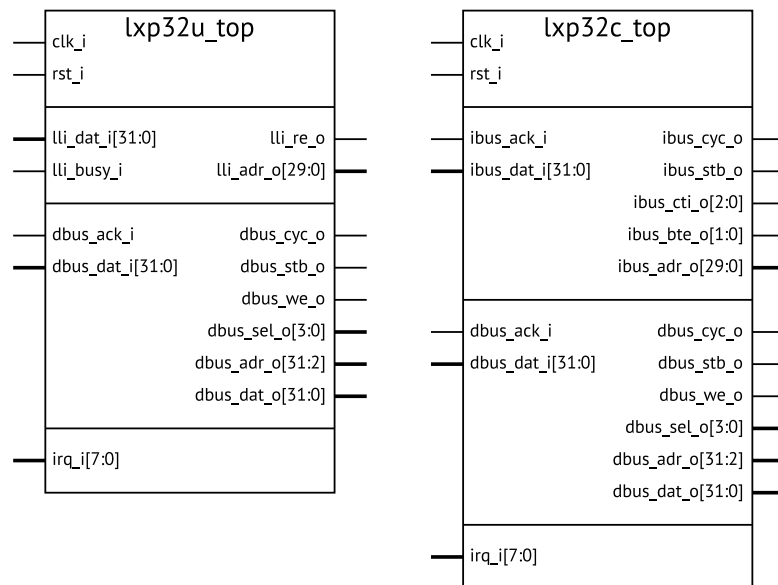


Figure 3.2: Schematic symbols for LXP32U and LXP32C

recommended for use in situations where LLI combinatorial delays are unacceptable.

Both LXP32U and LXP32C use the WISHBONE protocol for the data bus.

3.2 Ports

Port	Direction	Bus width	Description
<i>Global signals</i>			
clk_i	in	1	System clock
rst_i	in	1	Synchronous reset, active high
<i>Instruction bus – Low Latency Interface (LXP32U only)</i>			
lli_re_o	out	1	Read enable output, active high
lli_adr_o	out	30	Address output
lli_dat_i	in	32	Data input
lli_busy_i	in	1	Busy flag input, active high
<i>Instruction bus – WISHBONE (LXP32C only)</i>			
ibus_cyc_o	out	1	Cycle output
ibus_stb_o	out	1	Strobe output
ibus_cti_o	out	3	Cycle type identifier
ibus_bte_o	out	2	Burst type extension
ibus_ack_i	in	1	Acknowledge input
ibus_adr_o	out	30	Address output
ibus_dat_i	in	32	Data input
<i>Data bus</i>			
dbus_cyc_o	out	1	Cycle output
dbus_stb_o	out	1	Strobe output
dbus_we_o	out	1	Write enable output
dbus_sel_o	out	4	Select output
dbus_ack_i	in	1	Acknowledge input
dbus_adr_o	out	30	Address output
dbus_dat_o	out	32	Data output
dbus_dat_i	in	32	Data input
<i>Other ports</i>			
irq_i	in	8	Interrupt requests

3.3 Generics

The following generics can be used to configure the LXP32 IP core.

DBUS_RMW

This boolean generic controls how byte-granular data bus access is performed. LXP32 byte-granular instructions are **lub** (*Load Unsigned Byte*), **lsb** (*Load Signed Byte*) and **sb** (*Store Byte*).

Some WISHBONE buses provide hardware support for byte-granular access by the means of the [SEL_I()] (byte enable) port. In this case this option should be left to `false`, which is the default value.

LXP32 can also emulate byte-granular transactions. This works even if the data bus doesn't support byte-granular access natively. To enable this feature, `DBUS_RMW` should be set to `true`. The emulation works like this:

- When executing the **sb** instruction, the CPU performs an RMW (read-modify-write) cycle, reading the entire word, replacing a byte in it and writing the word back.
- When executing the **lub** and **lsb** instructions, the CPU reads the entire word and extracts the requested byte.

Either way, when this option is set to `true`, `dbus_sel_o` is tied to "1111".

DIVIDER_EN

LXP32 includes a divider unit which has quite a low performance but occupies a considerable amount of resources. It can be disabled by setting this boolean generic to `false`.

IBUS_BURST_SIZE

This integer generic determines the instruction bus burst size. Default value is 16. Only for LXP32C.

IBUS_PREFETCH_SIZE

This integer generic determines the number of words that the instruction cache will read ahead from the current instruction pointer. Default value is 32. Only for LXP32C.

MUL_ARCH

This string generic selects the multiplier hardware implementation. LXP32 provides three multiplier options:

- "dsp" is the fastest architecture designed for technologies that provide fast parallel 16×16 multipliers, which includes most modern FPGA families. One multiplication takes 2 clock cycles.
- "opt" architecture uses a semi-parallel multiplication algorithm based on carry-save accumulation of partial products. It is designed for technologies that do not provide fast 16×16 multipliers. One multiplication takes 6 clock cycles.
- "seq" is a fully sequential design. One multiplication takes 34 clock cycles.

The default multiplier architecture is "dsp". This option is recommended for most modern FPGA devices regardless of optimization goal since it is not only the fastest, but also occupies the least amount of general-purpose logic resources. However, it will create a timing bottleneck on technologies that lack fast multipliers.

For older FPGA families that don't provide dedicated multipliers the "opt" architecture can be used if decent throughput is still needed. It is designed to avoid creating a timing bottleneck on such technologies. Alternatively, "seq" architecture can be used when throughput is not a concern.

START_ADDR

Address of the first instruction to be executed after CPU reset, represented as `std_logic_vector`. Default value is 0. The two least significant bits are ignored as instructions are always word-aligned.

3.4 Clock and reset

All flip-flops in the CPU are triggered by a rising edge of the `clk_i` signal. No specific requirements are imposed on the `clk_i` signal apart from usual constraints on setup and hold times.

LXP32 is reset synchronously when the `rst_i` signal is asserted. If the system reset signal comes from an asynchronous source, a synchronization circuit must be used; an example of such a circuit is shown on Figure 3.3.

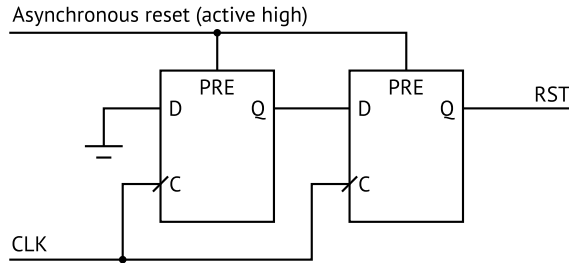


Figure 3.3: Reset synchronization circuit

In SRAM-based FPGAs flip-flops and RAM blocks have deterministic state after a bitstream is loaded. On such technologies LXP32 can operate without reset. In this case the `rst_i` port can be tied to a logical 0 in the RTL design to allow the synthesizer to remove redundant logic.

`clk_i` and `rst_i` signals also serve the role of `[CLK_I]` and `[RST_I]` WISHBONE signals, respectively, for both instruction and data buses.

3.5 Low Latency Interface

Low Latency Interface (LLI) is a simple pipelined synchronous protocol with a typical latency of 1 cycle used by LXP32U to fetch instructions. It was designed to allow simple connection of the CPU to on-chip program RAM or cache. The timing diagram of the LLI is shown on Figure 3.4.

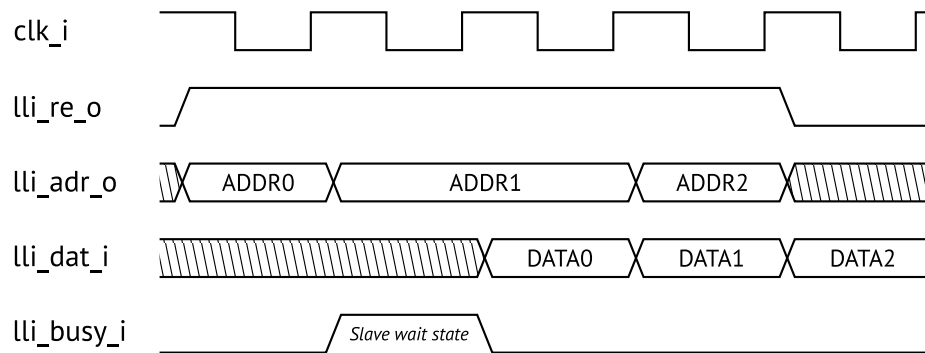


Figure 3.4: Low Latency Interface timing diagram (LXP32U)

To request a word, the master produces its address on `lli_adr_o` and asserts `lli_re_o`. The request is considered valid when `lli_re_o` is high and `lli_busy_i` is low on the same clock cycle. On the next cycle after a valid request, the slave must either produce data on `lli_dat_i` or assert `lli_busy_i` to indicate that data are not ready. `lli_busy_i` must be held high until the valid data are present on the `lli_dat_i` port.

The data provided by the slave are only required to be valid on the next cycle after a valid request (if `lli_busy_i` is not asserted) or on the cycle when `lli_busy_i` is deasserted after being held high. Otherwise `lli_dat_i` is undefined.

The values of `lli_re_o` and `lli_adr_o` are not guaranteed to be preserved by the master while the slave is busy.

The simplest slaves such as on-chip RAM blocks which are never busy can be trivially connected to the LLI by connecting address, data and read enable ports and tying the `lli_busy_i` signal to a logical 0 (you can even ignore `lli_re_o` in this case, although doing so can theoretically increase power consumption).

Since the `lli_re_o` output signal is not registered, this interface is not suitable for interaction with off-chip peripherals. Also, care should be taken to avoid introducing too much additional combinatorial delay on its outputs.

The instruction bus, whether LLI or WISHBONE, doesn't support access to individual bytes and uses a 30-bit address port to address 32-bit words (instructions are always word-aligned). The lower two bits of the 32-bit address are ignored for the purpose of addressing. Consider the following example:

```
lc r0, 0x10000000
jmp r0
// 0x04000000 will appear on lli_adr_o or ibus_adr_o
```

3.6 WISHBONE instruction bus

The LXP32C CPU fetches instructions over the WISHBONE bus. Its parameters are defined in the WISHBONE datasheet (Appendix D). For a detailed description of the bus protocol refer to the WISHBONE specification, revision B3.

With classic WISHBONE handshake decent throughput can be only achieved when the slave is able to terminate cycles asynchronously. It is

usually possible only for the simplest slaves which should probably be using the Low Latency Interface instead. To maximize throughput for complex, high latency slaves, LXP32C instruction bus uses optional WISHBONE address tags [CTI_OO] (Cycle Type Identifier) and [BTE_OO] (Burst Type Extension). These signals are hints allowing the slave to predict the address that will be set by the master in the next cycle and prepare data in advance. The slave can ignore these hints, processing requests as classic WISHBONE cycles, although performance would almost certainly suffer in this case.

A typical LXP32C instruction bus burst timing diagram is shown on Figure 3.5.

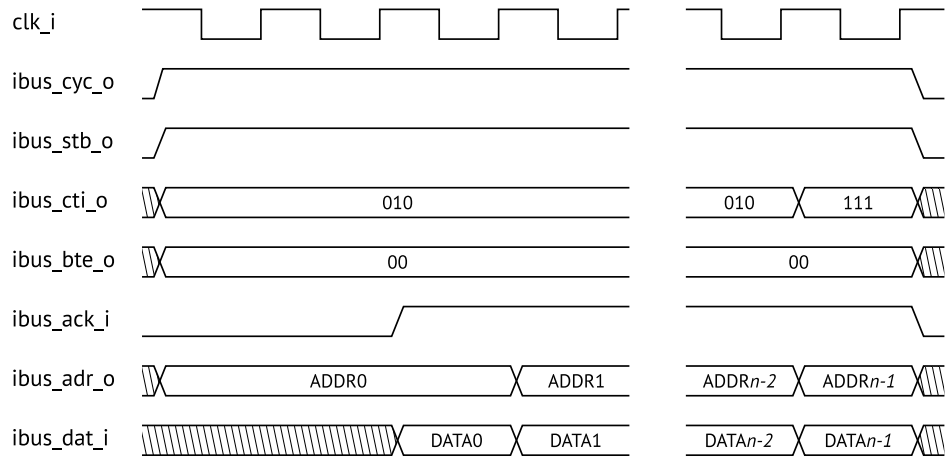


Figure 3.5: Typical WISHBONE instruction bus burst (LXP32C)

3.7 WISHBONE data bus

LXP32 uses the WISHBONE bus to interact with data memory and other peripherals. This bus is distinct from the instruction bus; its parameters are defined in the WISHBONE datasheet (Appendix D).

The data bus uses a 30-bit `dbus_adr_o` port to address 32-bit words; the `dbus_sel_o` port is used to select individual bytes to be written or read. The upper 30 bits of the address appear on the `dbus_adr_o` port. `dbus_sel_o` is set to "1111" when reading or writing whole words; for byte-granular transactions the lower two bits are decoded to create a 4-bit `dbus_sel_o` signal. Consider:

```

lc r0, 0x20000002
sb r0, 0x55
// write 0x55 to the address in r0
// 0x08000000 will appear on dbus_adr_o
// 0x4 will appear on dbus_sel_o

```

The byte-granular access feature is optional. If it is not needed, the `dbus_sel_o` port can be left unconnected. It is also possible to set the `DBUS_RMW` generic to `true` to enable byte-granular access emulation using the read-modify-write (RMW) cycle, which works even if the interconnect or slave doesn't provide the `[SEL_I()]` port (Section 3.3).

For a detailed description of the bus protocol refer to the WISHBONE specification, revision B3.

Typical timing diagrams for write and read cycles are shown on Figure 3.6. In these examples the peripheral terminates the cycle asynchronously; however, it can also introduce wait states by delaying the `dbus_ack_i` signal.

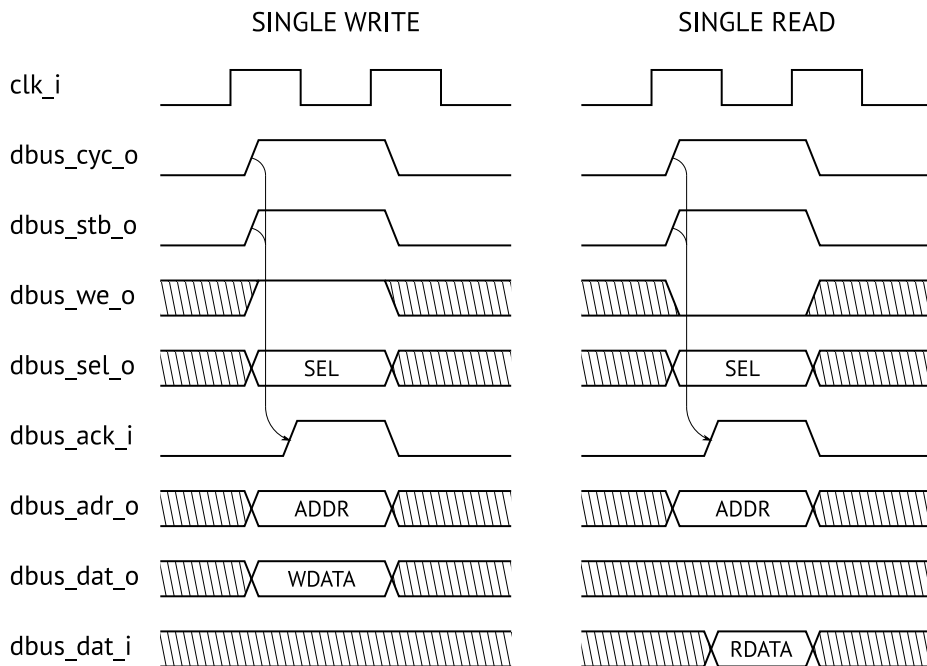


Figure 3.6: Typical WISHBONE data bus WRITE and READ cycles

3.8 Interrupts

All interrupt request signals must be synchronous with the system clock (`clk_i`); if coming from an asynchronous source, they must be synchronized using a sequence of at least two flip-flops clocked by `clk_i`. These flip-flops are not included in the LXP32 core in order not to increase interrupt processing delay for interrupt sources that are inherently synchronous. Failure to properly synchronize interrupt request signals will cause timing violations that will manifest itself as intermittent, hard to debug faults.

3.9 Synthesis and optimization

Technology specific primitives

LXP32 RTL design is described in behavioral VHDL. However, it can also benefit from certain special resources provided by most FPGA devices, namely, RAM blocks and dedicated multipliers. For improved portability, hardware description that can potentially be mapped to such resources is localized in separate design units:

- `lxp32_ram256x32` – a dual-port synchronous 256×32 bit RAM with one write port and one read port;
- `lxp32_mul16x16` – an unsigned 16×16 multiplier with an output register.

These design units contain behavioral description of respective hardware that is recognizable by FPGA synthesis tools. Usually no adjustments are needed as the synthesizer will automatically infer an appropriate primitive from its behavioral description. If automatic inference produces unsatisfactory results, these design units can be replaced with library element wrappers. The same is true for ASIC logic synthesis software which is unlikely to infer complex primitives.

LXP32 implements its own bypass logic dealing with situations when RAM read and write addresses collide. It does not depend on the read/write conflict resolution behavior of the underlying primitive.

General optimization guidelines

This subsection contains general advice on achieving satisfactory synthesis results regardless of the optimization goal. Some of these suggestions are also mentioned in other parts of this manual.

1. If the technology doesn't provide dedicated multiplier resources, consider using "opt" or "seq" multiplier architecture (Section 3.3).
2. Ensure that the instruction bus has adequate throughput. For LXP32C, check that the slave supports the WISHBONE registered feedback signals [CTI_I()] and [BTE_I()].
3. Multiplexing instruction and data buses, or connecting them to the same interconnect that allows only one master at a time to be active (i.e. *shared bus* interconnect topology) is not recommended. If you absolutely must do so, assign a higher priority level to the data bus, otherwise instruction prefetches will massively slow down data transactions.
4. For small programs, consider mapping code and data memory to the beginning or end of the address space (i.e. 0x00000000–0x000FFFFF or 0xFFFF0000–0xFFFFFFFF) to be able to load pointers with the **lcs** instruction which saves both memory and CPU cycles as compared to **lc**.

Optimizing for timing

1. Set up reasonable timing constraints. Do not overconstrain the design by more than 10–15 %.
2. Analyze the worst path. The natural LXP32 timing bottleneck usually goes from the scratchpad (register file) output through the ALU (in the Execute stage) to the scratchpad input. If timing analysis lists other critical paths, the problem can lie elsewhere. If the **rst_i** signal becomes a bottleneck, promote it to a global network or, with SRAM-based FPGAs, consider operating without reset (see Section 3.4). Critical paths affecting the WISHBONE state machines could indicate problems with interconnect performance.
3. Configure the synthesis tool to reduce the fanout limit. Note that setting this limit to a too small value can lead to an opposite effect.
4. Synthesis tools can support additional options to improve timing, such as the *Retiming* algorithm which rearranges registers and combinatorial logic across the pipeline in attempt to balance delays. The efficiency of such algorithms is not very predictable. In general, sloppy designs are the most likely to benefit from it, while for a carefully designed circuit timing can sometimes get worse.

Optimizing for area

1. Consider disabling the divider if not using it (see Section 3.3).
2. Relaxing timing constraints can sometimes allow the synthesizer to produce a more area-efficient circuit.
3. Increase the fanout limit in the synthesizer settings to reduce buffer replication.

Chapter 4

Hardware architecture

The LXP32 CPU is based on a 3-stage hazard-free pipelined architecture and uses a large RAM-based register file (scratchpad) with two read ports and one write port. The pipeline includes the following stages:

- *Fetch* – fetches instructions from the program memory.
- *Decode* – decodes instructions and reads register operand values from the scratchpad.
- *Execute* – executes instructions and writes the results (if any) to the scratchpad.

LXP32 instructions are encoded in such a way that operand register numbers can be known without decoding the instruction (Section 2.2). When the *Fetch* stage produces an instruction, scratchpad input addresses are set immediately, before the instruction itself is decoded. If the instruction does not use one or both of the register operands, the corresponding data read from the scratchpad are discarded. Collision bypass logic in the scratchpad detects situations where the *Decode* stage tries to read a register which is currently being written by the *Execute* stage and forwards its value, bypassing the RAM block and avoiding Read After Write (RAW) pipeline hazards. Other types of data hazards are also impossible with this architecture.

As an example, consider the following simple code chunk:

```
mov r0, 10 // alias for add r0, 10, 0
mov r1, 20 // alias for add r1, 20, 0
add r2, r0, r1
```

Table 4.1 illustrates how this chunk is processed by the LXP32 pipeline. Note that on the fourth cycle the *Decode* stage requests the r1 register value

while the *Execute* stage writes to the same register. Collision bypass logic in the scratchpad ensures that the *Decode* stage reads the correct (new) value of *r1* without stalling the pipeline.

Table 4.1: Example of the LXP32 pipeline operation

Cycle	Fetch	Decode	Execute
1	add <i>r0</i> , 10, 0		
2	add <i>r1</i> , 20, 0	add <i>r0</i> , 10, 0 Request <i>r10</i> (discarded) Request <i>r0</i> (discarded) Pass 10 and 0 as operands	
3	add <i>r2</i> , <i>r0</i> , <i>r1</i>	add <i>r1</i> , 20, 0 Request <i>r20</i> (discarded) Request <i>r0</i> (discarded) Pass 20 and 0 as operands	Perform the addition Write 10 to <i>r0</i>
4		add <i>r2</i> , <i>r0</i> , <i>r1</i> Request <i>r0</i> Request <i>r1</i> (bypass) Pass 10 and 20 as operands	Perform the addition Write 20 to <i>r1</i>
5			Perform the addition Write 30 to <i>r2</i>

When an instruction takes more than one cycle to execute, the *Execute* stage simply stalls the pipeline.

Branch hazards are impossible in LXP32 as well since the pipeline is flushed whenever an execution transfer occurs.

Chapter 5

Simulation

LXP32 package includes an automated verification environment (self-checking testbench) which verifies the LXP32 CPU functional correctness. The environment consists of two major parts: a test platform which is a SoC-like design providing peripherals for the CPU to interact with, and the testbench itself which loads test firmware and monitors the platform's output signals. Like the CPU itself, the test environment is written in VHDL-93.

A separate testbench for the instruction cache (`lxp32_icache`) is also provided. It can be invoked similarly to the main CPU testbench.

5.1 Requirements

The following software is required to simulate the LXP32 design:

- An HDL simulator supporting VHDL-93. LXP32 package includes scripts (makefiles) for the following simulators:
 - GHDL – a free and open source VHDL simulator¹;
 - NVC – another free and open source VHDL simulator²;
 - Mentor Graphics® ModelSim® simulator (`vsim`);
 - Xilinx® Vivado® Simulator (`xsim`).

With GHDL and NVC, a waveform viewer such as GTKWave is also recommended (Figure 5.1)³.

¹<http://ghdl.free.fr/>

²<https://www.nickg.me.uk/nvc/>

³<http://gtkwave.sourceforge.net/>

Other simulators can be used with some preparations (Section 5.3).

- GNU make and coreutils are needed to simulate the design using the provided makefiles. Under Microsoft® Windows®, MSYS or Cygwin can be used.
- LXP32 assembler/linker program (lxp32asm) must be present (Section 6.1). A prebuilt executable for Microsoft® Windows® is already included in the LXP32 package, for other operating systems lxp32asm must be built from source (Section 6.4).

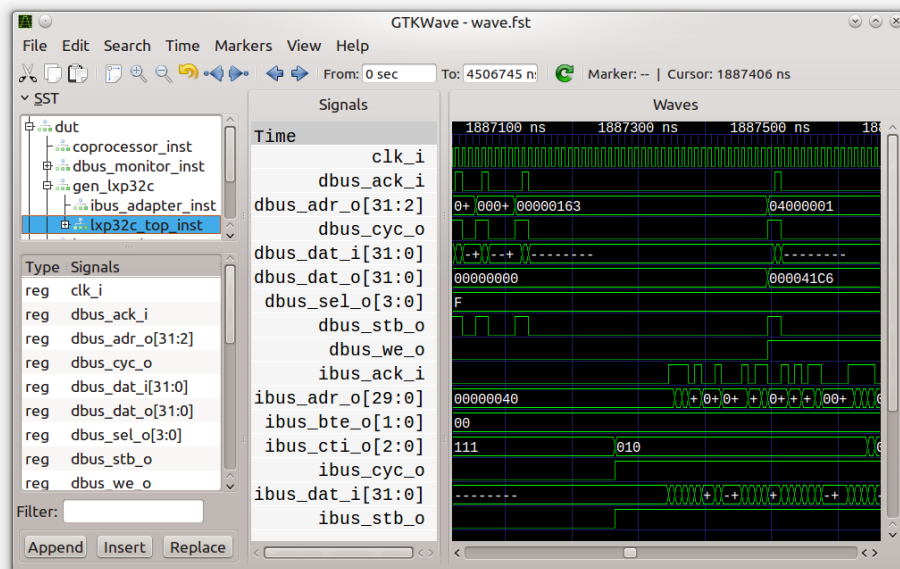


Figure 5.1: GTKWave displaying the LXP32 waveform dump produced by GHDL

5.2 Running simulation using makefiles

To simulate the design, go to the `verify/lxp32/run/<simulator>` directory and run `make`. The following make targets are supported:

- `batch` – simulate the design in batch mode. Results will be written to the standard output. This is the default target.

- `gui` – simulate the design in GUI mode. Note: with GHDL and NVC, the simulation itself will be run in batch mode; upon a completion, GTKWave will be run automatically to display the dumped waveforms.
- `compile` – compile only, don't run simulation.
- `clean` – delete all the produced artifacts.

5.3 Running simulation manually

LXP32 testbench can be also run manually. The following steps must be performed:

1. Compile the test firmware in the `verify/lxp32/src/firmware` directory:

```
lxp32asm -f textio filename.asm -o filename.ram
```

Produced `*.ram` files must be placed to the simulator's working directory.

2. Compile the LXP32 RTL description (`rtl` directory).
3. Compile the common package (`verify/common_pkg`).
4. Compile the test platform (`verify/lxp32/src/platform` directory).
5. Compile the testbench itself (`verify/lxp32/src/tb` directory).
6. Simulate the `tb` design unit defined in the `tb.vhd` file.

It shouldn't be hard to create a script for a new simulator using one of the existing makefiles as an example.

5.4 Testbench parameters

Simulation parameters can be configured by overriding generics defined by the `tb` design unit:

- `CPU_DBUS_RMW` – `DBUS_RMW` CPU generic value (see Section 3.3). Default is `false`.
- `CPU_MUL_ARCH` – `MUL_ARCH` CPU generic value (see Section 3.3). Default is `"dsp"`.

- `MODEL_LXP32C` – simulate the LXP32C version. Default is `true`. If set to `false`, LXP32U is simulated instead.
- `TEST_CASE` – if set to "all" (default), all tests are executed. Otherwise, specifies the test case to run.
- `THROTTLE_DBUS` – perform pseudo-random data bus throttling. Default is `true`.
- `THROTTLE_IBUS` – perform pseudo-random instruction bus throttling. Default is `true`.
- `VERBOSE` – print more messages. Default is `false`.

These generics can be also set using make parameters (currently works only for `ghdl`, `nvc` and `vsim`), for example:

```
make MODEL_LXP32C=false TEST_CASE=test001
```

Chapter 6

Development tools

6.1 `lxp32asm` – Assembler and linker

`lxp32asm` is a combined assembler and linker for the LXP32 platform. It takes one or more input files and produces executable code for the CPU. Input files can be either source files in the LXP32 assembly language (Appendix C) or *linkable objects*. Linkable object is a relocatable format for storing compiled LXP32 code together with symbol information.

`lxp32asm` operates in two stages:

1. Compile.

Source files are compiled to linkable objects.

2. Link.

Linkable objects are combined into a single executable module. References to symbols defined in external modules are resolved at this stage.

In the simplest case there is only one input source file which doesn't contain external symbol references. If there are multiple input files, one of them must define the entry (or Entry) symbol at the beginning of the code.

Command line syntax

```
lxp32asm [ options | input files ]
```

General options

- `-c` – compile only (skip the Link stage).
- `-h, --help` – display a short help message and exit.
- `-o file` – output file name.
- `--` – do not interpret the subsequent command line arguments as options. Can be used if there are input file names starting with a dash.

Compiler options

- `-i dir` – add *dir* to the list of directories used to search for included files. Multiple directories can be specified with multiple `-i` arguments.

Linker options (ignored in compile-only mode)

- `-a align` – object alignment. Must be a power of 2 and can't be less than 4. Default value is 4.
- `-b addr` – base address, that is, the address in memory where the executable image will be located. Must be a multiple of object alignment. Default value is 0.
- `-f fmt` – executable image format. See below for the list of supported formats.
- `-m file` – generate a map file. A map file is a human-readable list of all object and symbol addresses in the executable image.
- `-s size` – size of the executable image. Must be a multiple of 4. If total code size is less than the specified value, the executable image is padded with zeros. By default, the image is not padded.

Output formats

Output formats that can be specified with the `-f` command line option are listed below.

- `bin` – raw binary image (little-endian). This is the default format.
- `textio` – text format representing binary data as a sequence of zeros and ones. This format can be directly read from VHDL (using the `std.textio` package) or Verilog® (using the `$readmemb` function).

- `dec` – text format representing each word as a decimal number.
- `hex` – text format representing each word as a hexadecimal number.

6.2 `lxp32dump` – Disassembler

`lxp32dump` takes an executable image and produces a source file in LXP32 assembly language. The produced file is a valid program that can be compiled by `lxp32asm`.

Command line syntax

```
lxp32dump [ options | input file ]
```

Supported options are:

- `-b addr` – executable image base address, only used for comments.
- `-f fmt` – input file format. All `lxp32asm` output formats are supported. If this option is not supplied, autodetection is performed.
- `-h, --help` – display a short help message and exit.
- `-na` – do not use instruction aliases (such as `mov`, `ret`, `not`) and register aliases (such as `sp`, `rp`).
- `-o file` – output file name. By default, the standard output stream is used.
- `--` – do not interpret subsequent command line arguments as options.

6.3 `wigen` – Interconnect generator

`wigen` is a small tool that generates VHDL description of a simple WISHBONE interconnect based on shared bus topology. It supports any number of masters and slaves. The interconnect can then be used to create a SoC based on LXP32.

For interconnects with multiple masters a priority-based arbitration circuit is inserted with lower-numbered masters taking precedence. However, when a bus cycle is in progress (`[CYC_O]` is asserted by the active master), the arbiter will not interrupt it even if a master with a higher priority level requests bus ownership.

Command line syntax

wigen [*option(s)*] *nm ns ma sa ps* [*pg*]

- *nm* – number of masters,
- *ns* – number of slaves,
- *ma* – master address width,
- *sa* – slave address width,
- *ps* – port size (8, 16, 32 or 64),
- *pg* – port granularity (8, 16, 32 or 64, default: the same as port size).

Supported options are:

- *-e entity* – name of the design entity (default is "intercon").
- *-h, --help* – display a short help message and exit.
- *-o file* – output file name (default is *entity.vhd*).
- *-p* – generate pipelined arbiter (reduced combinatorial delays, increased latency).
- *-r* – generate WISHBONE registered feedback signals ([CTI_IO()] and [BTE_IO()]).
- *-u* – generate unsafe slave decoder (reduced combinatorial delays and resource usage, may not work properly if the address is invalid).

6.4 Building from source

Prebuilt tool executables for 32-bit Microsoft® Windows® are included in the LXP32 IP core package. For other platforms the tools must be built from source. Since they are developed in C++ using only the standard library, it should be possible to build them for any platform that provides a modern C++ compiler.

Requirements

The following software is required to build LXP32 tools from source:

1. A modern C++ compiler, such as Microsoft® Visual Studio® 2013 or newer, GCC 4.8 or newer, Clang 3.4 or newer.
2. CMake 3.3 or newer.

Build procedure

This software uses CMake as a build system generator. Building it involves two steps: first, the cmake program is invoked to generate a native build environment (a set of Makefiles or an IDE project); second, the generated environment is used to build the software. More details can be found in the CMake documentation.

Examples

In the following examples, it is assumed that the commands are run from the tools subdirectory of the LXP32 IP core package tree.

For Microsoft® Visual Studio®:

```
mkdir build
cd build
cmake -G "NMake Makefiles" ../src
nmake
nmake install
```

For MSYS:

```
mkdir build
cd build
cmake -G "MSYS Makefiles" ../src
make
make install
```

For MinGW without MSYS:

```
mkdir build
cd build
cmake -G "MinGW Makefiles" ../src
mingw32-make
mingw32-make install
```

For other platforms:

```
mkdir build  
cd build  
cmake ../src  
make  
make install
```

Appendix A

Instruction set reference

See Section 2.2 for a general description of LXP32 instruction encoding.

A.1 List of instructions by group

Instruction	Description	Opcode
<i>Data transfer</i>		
mov	Move	alias for add dst, src, 0
lc	Load Constant	000001
lcs	Load Constant Short	101xxx
lw	Load Word	001000
lub	Load Unsigned Byte	001010
lsb	Load Signed Byte	001011
sw	Store Word	001100
sb	Store Byte	001110
<i>Arithmetic operations</i>		
add	Add	010000
sub	Subtract	010001
neg	Negate	alias for sub dst, 0, src
mul	Multiply	010010
divu	Divide Unsigned	010100
divs	Divide Signed	010101
modu	Modulo Unsigned	010110
mods	Modulo Signed	010111

<i>Logical operations</i>		
not	Bitwise Not	alias for xor dst, src, -1
and	Bitwise And	011000
or	Bitwise Or	011001
xor	Bitwise Exclusive Or	011010
sl	Shift Left	011100
sru	Shift Right Unsigned	011110
srs	Shift Right Signed	011111
<i>Execution transfer</i>		
jmp	Jump	100000
cjmpxxx	Compare and Jump	11xxxx (xxxx = condition)
call	Call Procedure	alias for xcall rp, addr
xcall	Extended Call	100001
ret	Return from Procedure	alias for jmp rp
iret	Interrupt Return	alias for jmp irp
<i>Miscellaneous instructions</i>		
nop	No Operation	000000
hlt	Halt	000010

A.2 Alphabetical list of instructions

add – Add

Syntax

add DST, RD1, RD2

Encoding

010000 T1 T2 DST RD1 RD2

Example: **add** r2, r1, 10 → 0x4202010A

Operation

DST := RD1 + RD2

and – Bitwise And**Syntax****and** DST, RD1, RD2**Encoding**

011000 T1 T2 DST RD1 RD2

Example: **and** r2, r1, 0x3F → 0x6202013F**Operation**DST := RD1 $\wedge$ RD2**call – Call Procedure****Syntax****call** RD1Alias for **xcall** rp, RD1**cjmpxxx – Compare and Jump**

Compare two operands and transfer execution to the specified address if a condition is satisfied.

Syntax**cjmpe** DST, RD1, RD2 (Equal)**cjmpne** DST, RD1, RD2 (Not Equal)**cjmpsg** DST, RD1, RD2 (Signed Greater)**cjmpsge** DST, RD1, RD2 (Signed Greater or Equal)**cjmpsl** DST, RD1, RD2 (Signed Less)**cjmpsle** DST, RD1, RD2 (Signed Less or Equal)**cjmpug** DST, RD1, RD2 (Unsigned Greater)**cjmpuge** DST, RD1, RD2 (Unsigned Greater or Equal)**cjmpul** DST, RD1, RD2 (Unsigned Less)**cjmpule** DST, RD1, RD2 (Unsigned Less or Equal)

Encoding

OPCODE T1 T2 DST RD1 RD2

Opcodes:

cjmpe	111000
cjmpne	110100
cjmpsg	110001
cjmpsge	111001
cjmpug	110010
cjmpuge	111010

cjmpsl, **cjmpsle**, **cjmpul**, **cjmpule** instructions are aliases for **cjmpsg**, **cjmpsge**, **cjmpug**, **cjmpuge**, respectively, with RD1 and RD2 operands swapped.

Example: **cjmpuge** r2, r1, 5 $\rightarrow$ 0xEA020105

Operation

if *condition* then goto DST

Pointer in DST is interpreted as described in Section 2.4. Unlike most instructions, **cjmpxxx** does not write to DST.

divs – Divide Signed**Syntax**

divs DST, RD1, RD2

Encoding

010101 T1 T2 DST RD1 RD2

Example: **divs** r2, r1, -3 $\rightarrow$ 0x560201FD

Operation

DST := (*signed*) RD1 / (*signed*) RD2

The result is rounded towards zero and is undefined if RD2 is zero. If the CPU was configured without a divider, this instruction returns 0.

divu – Divide Unsigned**Syntax****divu** DST, RD1, RD2**Encoding**

010100 T1 T2 DST RD1 RD2

Example: **divu** r2, r1, 73 → 0x52020107**Operation**

DST := RD1 / RD2

The result is rounded towards zero and is undefined if RD2 is zero. If the CPU was configured without a divider, this instruction returns 0.

hlt – Halt

Halt the CPU until an enabled interrupt is received.

Syntax**hlt****Encoding**

000010 0 0 00000000 00000000 00000000

Operation

Pause execution until an interrupt is received.

jmp – Jump

Transfer execution to the address pointed by the operand.

Syntax**jmp** RD1

Encoding

100000 1 0 00000000 RD1 00000000

RD1 must be a register.

Example: **jmp** r1 → 0x82000100

Operation

goto RD1

Pointer in RD1 is interpreted as described in Section 2.4.

iret – Interrupt Return

Return from an interrupt handler.

Syntax

iret

Alias for **jmp** irp.

lc – Load Constant

Load a 32-bit word to the specified register. Note that values from the [-1048576; 1048575] range can be loaded more efficiently using the **lcs** instruction.

Syntax

lc DST, WORD32

Encoding

000001 0 0 DST 00000000 00000000 WORD32

Unlike other instructions, **lc** occupies two 32-bit words.

Example: **lc** r1, 0x12345678 → 0x04010000 0x12345678

Operation

DST := WORD32

lcs – Load Constant Short

Load a signed value from the [-1048576; 1048575] range (a sign extended 21-bit value) to the specified register. Unlike the **lc** instruction, this instruction is encoded as a single word.

Syntax

lcs DST, VAL

Encoding

101 VAL[20:16] DST VAL[15:0]

Example: **lcs** r1, -1000000 → 0xB001BDC0

Operation

DST := (*signed*) VAL

lsb – Load Signed Byte

Load a byte from the specified address to the register, performing sign extension.

Syntax

lsb DST, RD1

Encoding

001011 1 0 DST RD1 00000000

RD1 must be a register.

Example: **lsb** r2, r1 → 0x2E020100

Operation

DST := (*signed*) (*(BYTE*)RD1)

Pointer in RD1 is interpreted as described in Section 2.4.

lub** – Load Unsigned Byte**

Load a byte from the specified address to the register. Higher 24 bits are zeroed.

Syntax

lub**** DST, RD1

Encoding

001010 1 0 DST RD1 00000000

RD1 must be a register.

Example: **l**ub**** r2, r1 → 0x2A020100

Operation

DST := *(BYTE*)RD1

Pointer in RD1 is interpreted as described in Section 2.4.

lw** – Load Word**

Load a word from the specified address to the register.

Syntax

lw**** DST, RD1

Encoding

001000 1 0 DST RD1 00000000

RD1 must be a register.

Example: **l**w**** r2, r1 → 0x22020100

Operation

DST := *RD1

Pointer in RD1 is interpreted as described in Section 2.4.

mods – Modulo Signed**Syntax**

mods DST, RD1, RD2

Encoding

010111 T1 T2 DST RD1 RD2

Example: **mods** r2, r1, 10 $\rightarrow$ 0x5E02010A

Operation

$DST := (\text{signed})\ RD1 \bmod (\text{signed})\ RD2$

Modulo operation satisfies the following condition: if $Q = A/B$ and $R = A \bmod B$, then $A = B \cdot Q + R$.

The result is undefined if RD2 is zero. If the CPU was configured without a divider, this instruction returns 0.

modu – Modulo Unsigned**Syntax**

modu DST, RD1, RD2

Encoding

010110 T1 T2 DST RD1 RD2

Example: **modu** r2, r1, 10 $\rightarrow$ 0x5A02010A

Operation

$DST := RD1 \bmod RD2$

Modulo operation satisfies the following condition: if $Q = A/B$ and $R = A \bmod B$, then $A = B \cdot Q + R$.

The result is undefined if RD2 is zero. If the CPU was configured without a divider, this instruction returns 0.

mov – Move**Syntax****mov** DST, RD1Alias for **add** DST, RD1, 0**mul – Multiply**

Multiply two 32-bit values. The result is also 32-bit.

Syntax**mul** DST, RD1, RD2**Encoding**

010010 T1 T2 DST RD1 RD2

Example: **mul** r2, r1, 3 → 0x4A020103**Operation**

DST := RD1 * RD2

Since the product width is the same as the operand width, the result of a multiplication does not depend on operand signedness.

neg – Negate**Syntax****neg** DST, RD2Alias for **sub** DST, 0, RD2**nop – No Operation****Syntax****nop****Encoding**

000000 0 0 00000000 00000000 00000000

Operation

This instruction does not alter the machine state.

not – Bitwise Not**Syntax**

not DST, RD1

Alias for **xor** DST, RD1, -1.

or – Bitwise Or**Syntax**

or DST, RD1, RD2

Encoding

011001 T1 T2 DST RD1 RD2

Example: **or** r2, r1, 0x3F → 0x6602013F

Operation

DST := RD1 $\vee$ RD2

ret – Return from Procedure

Return from a procedure.

Syntax

ret

Alias for **jmp** rp.

sb – Store Byte

Store the lowest byte from the register to the specified address.

Syntax

sb RD1, RD2

Encoding

001110 1 T2 00000000 RD1 RD2

RD1 must be a register.

Example: **sb** r2, r1 → 0x3B000201

Operation

$*(\text{BYTE}^*)\text{RD1} := \text{RD2} \wedge 0\text{x}000000\text{FF}$

Pointer in RD1 is interpreted as described in Section 2.4.

sl – Shift Left**Syntax**

sl DST, RD1, RD2

Encoding

011100 T1 T2 DST RD1 RD2

Example: **sl** r2, r1, 5 → 0x72020105

Operation

$\text{DST} := \text{RD1} \ll \text{RD2}$

The result is undefined if RD2 is outside the [0; 31] range.

srs – Shift Right Signed**Syntax**

srs DST, RD1, RD2

Encoding

011111 T1 T2 DST RD1 RD2

Example: **srs** r2, r1, 5 → 0x7E020105

Operation

$\text{DST} := ((\text{signed}) \text{RD1}) \gg \text{RD2}$

The result is undefined if RD2 is outside the [0; 31] range.

sru – Shift Right Unsigned**Syntax**

sru DST, RD1, RD2

Encoding

011110 T1 T2 DST RD1 RD2

Example: **sru** r2, r1, 5 → 0x7A020105

Operation

DST := RD1 >> RD2

The result is undefined if RD2 is outside the [0; 31] range.

sub – Subtract**Syntax**

sub DST, RD1, RD2

Encoding

010001 T1 T2 DST RD1 RD2

Example: **sub** r2, r1, 5 → 0x46020105

Operation

DST := RD1 - RD2

sw – Store Word

Store the value of the register to the specified address.

Syntax

sw RD1, RD2

Encoding

001100 1 T2 00000000 RD1 RD2

RD1 must be a register.

Example: **sw** r2, r1 → 0x33000201

Operation

*RD1 := RD2

Pointer in RD1 is interpreted as described in Section 2.4.

xcall – Extended Call

Save a pointer to the next instruction in the register specified as the first operand and transfer execution to the address pointed by the second operand.

Syntax

xcall DST, RD1

Encoding

100001 1 0 DST RD1 00000000

RD1 must be a register.

Example: **xcall** rp, r1 → 0x86FE0100

Operation

DST := *return_address*

goto RD1

Pointer in RD1 is interpreted as described in Section 2.4. **xcall** rp, RD1 is equivalent to **call** RD1.

xor – Bitwise Exclusive Or**Syntax**

xor DST, RD1, RD2

Encoding

011010 T1 T2 DST RD1 RD2

Example: **xor** r2, r1, 0x3F $\rightarrow$ 0x6A02013F

Operation

DST := RD1 $\oplus$ RD2

Appendix B

Instruction cycle counts

Cycle counts for LXP32 instructions are listed in Table B.1. These data are provided for reference purposes; the software should not depend on them as they can change in future hardware revisions.

Table B.1: Instruction cycle counts

Instruction	Cycles	Instruction	Cycles
add	1	mov	1
and	1	mul	2, 6 or 34 ³
call	4	neg	1
cjmpxxx	5 or 2 ¹	nop	1
divs	36	not	1
divu	36	or	1
hlt	N/A	ret	4
jmp	4	sb	$\geq 2^2$
iret	4	sl	2
lc	2	srs	2
lcs	1	sru	2
lsb	$\geq 3^2$	sub	1
lub	$\geq 3^2$	sw	$\geq 2^2$
lw	$\geq 3^2$	xcall	4
mods	37	xor	1
modu	37		

¹Depends on whether the jump is taken or not.

²Depends on the data bus latency.

³Depends on the multiplier architecture. See Section 3.3.

Appendix C

LXP32 assembly language

This appendix defines the assembly language used by LXP32 development tools.

C.1 Comments

LXP32 assembly language supports C style comments that can span across multiple lines and single-line C++ style comments:

```
/*  
 * This is a comment.  
*/
```

```
// This is also a comment
```

From a parser's point of view comments are equivalent to whitespace.

C.2 Literals

LXP32 assembly language uses numeric and string literals similar to those provided by the C programming language.

Numeric literals can take form of decimal, hexadecimal or octal numbers. Literals prefixed with 0x are interpreted as hexadecimal, literals prefixed with 0 are interpreted as octal, other literals are interpreted as decimal. A numeric literal can also start with an unary plus or minus sign which is also considered a part of the literal.

String literals must be enclosed in double quotes. The most common escape sequences used in C are supported (Table C.1). Note that strings

are not null-terminated in the LXP32 assembly language; when required, terminating null character must be inserted explicitly.

Table C.1: Escape sequences used in string literals

Sequence	Interpretation
\\	Backslash character
\"	Double quotation mark
\'	Single quotation mark (can be also used directly)
\t	Tabulation character
\n	Line feed
\r	Carriage return
\xXX	Character with a hexadecimal code of XX (1–2 digits)
\XXX	Character with an octal code of XXX (1–3 digits)

C.3 Symbols

Symbols (labels) are used to refer to data or code locations. LXP32 assembly language does not have distinct code and data labels: symbols are used in both these contexts.

Symbol names must be valid identifiers. A valid identifier must start with an alphabetic character or an underscore, and may contain alphanumeric characters and underscores.

A symbol definition must be the first token in a source code line followed by a colon. A symbol definition can occupy a separate line (in which case it refers to the following statement). Alternatively, a statement can follow the symbol definition on the same line.

Symbols can be used as operands to the **lc** and **lcs** instruction statements. A symbol reference can end with a **@n** sequence, where *n* is a numeric literal; in this case it is interpreted as an offset (in bytes) relative to the symbol definition. For the **lcs** instruction, the resulting address must still fit into the sign extended 21-bit value range (0x00000000–0x000FFFFF or 0xFFFF0000–0xFFFFFFFF), otherwise the linker will report an error.

By default all symbols are local, that is, they can be only referenced from the module where they were defined. To make a symbol accessible from other modules, use the **#export** directive. To reference a symbol defined in another module use the **#import** directive.

A symbol named **entry** or **Entry** has a special meaning: it is used to inform the linker about the program entry point if there are multiple input

files. It does not have to be exported. If defined, this symbol must precede the first instruction or data definition statement in the module. Only one module in the program can define the entry symbol.

```

    lc r10, jump_label
    lc r11, data_word
// ...
    sw r11, r0 // store the value of r0 to the
                // location pointed by data_word
    jmp r10     // transfer execution to jump_label
// ...
jump_label:
    mov r1, r0
// ...
data_word:
    .word 0x12345678

```

C.4 Statements

Each statement occupies a single source code line. There are three kinds of statements:

- *Directives* provide directions for the assembler that do not directly cause code generation.
- *Data definition statements* insert arbitrary data to the generated code.
- *Instruction statements* insert LXP32 CPU instructions to the generated code.

Directives

The first token of a directive statement always starts with the # character.

```
#define identifier [ token ... ]
```

Defines a macro that will be substituted with zero or more tokens. The *identifier* must satisfy the requirements listed in Section C.3. Tokens can be anything, including keywords, identifiers, literals and separators (i.e. comma and colon characters).

#error [*msg*]

Raises a compiler error. If *msg* is supplied, uses it as an error message.

#export *identifier*

Declares *identifier* as an exported symbol. Exported symbols can be referenced by other modules.

#ifdef | **#ifndef** *identifier*

...

#else

...

#endif

Define C preprocessor-style conditional sections which are processed or not based on whether a certain macro has been defined. **#else** is optional. Can be nested.

#import *identifier*

Declares *identifier* as an imported symbol. Used to refer to symbols exported by other modules.

#include *filename*

Processes *filename* contents as it were literally inserted at the point of the **#include** directive. *filename* must be a string literal.

#message *msg*

Prints *msg* to the standard output stream. *msg* must be a string literal.

Data definition statements

The first token of a data definition statement always starts with the . (period) character.

.align [*alignment*]

Ensures that code generated by the next data definition or instruction statement is aligned to a multiple of *alignment* bytes, inserting padding zeros if needed. *alignment* must be a power of 2 and can't be less than 4. Default *alignment* is 4. Instructions and words are always at least word-aligned; the **.align** statement can be used to align them to a larger boundary, or to align byte data (see below).

The **.align** statement is not guaranteed to work if the requested alignment is greater than the section alignment specified for the linker (see Subsection 6.1).

.byte *token* [, *token* ...]

Inserts one or more bytes to the output code. Each *token* can be either a numeric literal with a valid range of [-128; 255] or a string literal. By default, bytes are not aligned.

To define a null-terminated string, the terminating null character must be inserted explicitly.

.reserve *n*

Inserts *n* zero bytes to the output code.

.word *token* [, *token* ...]

Inserts one or more 32-bit words to the output code. Tokens must be numeric literals.

Instruction statements

Instruction statements have the following general syntax:

instruction [*operand* [, *operand* ...]]

Depending on the instruction, operands can be registers, numeric literals or symbols. Supported instructions are listed in Appendix A.

Appendix D

WISHBONE datasheet

D.1 Instruction bus (LXP32C only)

<i>General information</i>	
WISHBONE revision	B3
Type of interface	MASTER
Supported cycles	BLOCK READ
<i>Signal names</i>	
clk_i	CLK_I
rst_i	RST_I
ibus_cyc_o	CYC_O
ibus_stb_o	STB_O
ibus_cti_o	CTI_O()
ibus_bte_o	BTE_O()
ibus_ack_i	ACK_I
ibus_adr_o	ADR_O()
ibus_dat_i	DAT_I()
<i>Supported tag signals</i>	
ibus_cti_o	Cycle Type Identifier (address tag) “010” (Incrementing burst cycle) “111” (End-of-Burst)
ibus_bte_o	Burst Type Extension (address tag) “00” (Linear burst)
<i>Dimensions</i>	
Port size	32

Port granularity	32
Maximum operand size	32
Data transfer ordering	BIG/LITTLE ENDIAN
Data transfer sequence	UNDEFINED

D.2 Data bus

<i>General information</i>	
WISHBONE revision	B3
Type of interface	MASTER
Supported cycles	SINGLE READ/WRITE RMW
<i>Signal names</i>	
clk_i	CLK_I
rst_i	RST_I
dbus_cyc_o	CYC_O
dbus_stb_o	STB_O
dbus_we_o	WE_O
dbus_sel_o	SEL_O()
dbus_ack_i	ACK_I
dbus_adr_o	ADR_O()
dbus_dat_o	DAT_O()
dbus_dat_i	DAT_I()
<i>Dimensions</i>	
Port size	32
Port granularity	8
Maximum operand size	32
Data transfer ordering	LITTLE ENDIAN
Data transfer sequence	UNDEFINED

Appendix E

List of changes

Version 1.5.1 (2026-09-06)

This release introduces a few testbench improvements. Support for a new simulator, NVC, has been added. It is also now possible to override testbench generics with make command-line arguments.

Version 1.5 (2026-09-03)

This release introduces support for a previously undocumented **xcall** instruction to the toolchain. The RTL description is unchanged.

Version 1.4 (2024-11-02)

This release introduces support for level triggered interrupts.

Version 1.3 (2022-08-28)

This release removes support for temporarily blocked interrupts (interrupts can still be disabled) and introduces wake-up interrupts.

Version 1.2 (2021-10-21)

This release introduces a few non-breaking changes to the software and testbench. The CPU RTL description hasn't been changed from the previous release.

- `lxp32asm` now supports C-style conditional processing directives: **`#ifdef`**, **`#ifndef`**, **`#else`** and **`#endif`**.
- **`#define`** directive can now declare a macro with zero substitute tokens.
- A new **`#error`** directive.
- Minor changes to the testbench.

Version 1.1 (2019-01-11)

This release introduces a minor but technically breaking hardware change: the `START_ADDR` generic, which used to be 30-bit, has been for convenience extended to a full 32-bit word; the two least significant bits are ignored.

The other breaking change affects the assembly language syntax. Previously all symbols used to be public, and multiple modules could not define symbols with the same name. As of now only symbols explicitly exported using the **`#export`** directive are public. **`#extern`** directive has been replaced by **`#import`**.

Other notable changes include:

- A new instruction, **`lcs`** (*Load Constant Short*), has been added, which loads a 21-bit sign extended constant to a register. Unlike **`lc`**, it is encoded as a single word and takes one cycle to execute.
- Optimizations in the divider unit. Division instructions (**`divs`** and **`divu`**) now take one fewer cycle to execute (modulo instructions are unaffected).
- LXP32 assembly language now supports a new instruction alias, **`neg`** (*Negate*), which is equivalent to **`sub dst, 0, src`**.

Version 1.0 (2016-02-20)

Initial public release.